Sabnaj Akter

Distributed Systems & Cloud Infrastructure Engineer | AppsCode Inc.

Codeforces Expert (1602) • **CodeChef 5-Star** • Distributed Systems & AI Infrastructure

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RESEARCH INTERESTS

Distributed Systems & Consensus, Fault Tolerance (Crash & Byzantine), Replication & Reliability, Cloud Computing & Infrastructure, Systems Security

EDUCATION

Khulna University of Engineering & Technology (KUET) **Khulna, Bangladesh**
BSc in Computer Science and Engineering (CGPA: 3.21/4.00) *Completed September 2025*

- **Relevant Coursework:** Operating Systems, Computer Networks, Database Systems, Data Structures, Algorithms, Machine Learning.

PUBLICATIONS & RESEARCH

BnLiteBait: Attention-Based Lightweight Clickbait Detection (Bengali)
Preprint (DOI: 10.13140/RG.2.2.31280.21766) *ResearchGate Link*

- Proposed a lightweight attention-based model for Bengali clickbait detection using headline–body fusion.
- Improved performance using ensemble bagging and t-SNE-based feature visualization analysis.

N-gram-Based Bengali Text Compression
Conference Paper — Under Review

- Developed an N-gram-based Bengali text compression framework using statistical language modeling.
- Built tokenization and dictionary generation pipeline and evaluated compression performance.

WORK EXPERIENCE

Software Engineer **2025 – Present**
AppsCode Inc. – Distributed Systems and Cloud-Native R&D *Dhaka, Bangladesh*

- Built cloud-native distributed database infrastructure on Kubernetes for stateful workloads, spanning provisioning, replication, and durable persistent storage.
- Researched and developed Kubernetes database operators for IBM DB2, DocumentDB, Milvus, and etcd.
- Implemented Raft-based leader election and failover, alongside disaster-recovery workflows designed to meet RPO/RTO targets and preserve data durability.
- Built a TCP-based cross-cluster disaster-recovery control plane and exercised its restore paths end to end under RPO/RTO targets.
- Built observability pipelines using Prometheus exporters, PromQL, and Grafana dashboards to collect failure and recovery telemetry.
- Characterized failure modes and recovery behavior from fault-injection telemetry across 1000+ workloads to localize durability-guarantee violations and resilience limits.

PROJECTS

KubeDB: IBM Db2 Operator [Code] [Docs]
Go, Kubernetes, Cloud, TCP, Controller Runtime, Prometheus, Grafana

- Implemented consensus-based failover and replication-aware recovery to ensure high availability and data durability under failure.
- Designed a TCP-based control plane for disaster recovery and cross-cluster communication.
- Built chaos / fault-injection test harnesses to measure recovery latency and validate resilience guarantees under 1000+ database workloads and multi-failure scenarios.
- Analyzed Prometheus metrics and Grafana dashboards to detect anomalies and failure signatures during recovery.
- Developed a Kubernetes operator for managing IBM Db2 database workloads in cloud-native environments.

KubeDB: DocumentDB Operator [Code] [Docs]
Go, Kubernetes, Raft, gRPC, Distributed Systems

- Designed distributed database orchestration workflows for DocumentDB clusters on Kubernetes.
- Implemented Raft-based consensus and leader election mechanisms for high availability and fault tolerance.
- Designed gRPC-based internal services for cluster coordination and inter-node communication.
- Worked on automated provisioning, failover orchestration, and distributed database lifecycle management.
- Built reliability and recovery workflows for resilient distributed database operations.
- Profiled failover recovery latency and durability-guarantee boundaries under large-scale fault injection and multi-failure scenarios.

KubeDB: Milvus Operator [Code]
Go, Kubernetes, etcd, Milvus (vector database), object storage, Multus, SR-IOV, GPU, fault injection

- Built the Kubernetes operator for Milvus and designed its physical base backup and point-in-time recovery: a consistent crash image from an etcd range dump, a continuous etcd watch log, and a versioned object-store pool, restored into a fresh cluster.
- Validated by fault injection (killed backup pods, partitions, etcd compaction) and by independently regenerating seeded vectors and comparing restored data within a 1e-5 tolerance, with negative controls; found and fixed silent-success defects, e.g. a PITR request ignored yet reported *Succeeded*, and a write fence leaked when a backup pod was killed.
- Implemented and tested GPU and SR-IOV (secondary-NIC, Multus) support for distributed Milvus clusters.

TECHNICAL SKILLS

Systems & Cloud Infrastructure: Kubernetes, Docker, Linux, Kubernetes Operators, CRDs, Raft
Networking, Backend & Observability: TCP, HTTP, gRPC, REST APIs, TLS, mTLS, Prometheus, PromQL, Grafana
Databases & Storage: IBM Db2, PostgreSQL, DocumentDB (document store), Milvus (vector database), etcd; replication, durability / RPO-RTO, disaster recovery, persistent volumes
Programming: Go, Python, C/C++, Java, SQL, Bash
Machine Learning & Data Methods: Attention-Based Models, Statistical Language Modeling (N-gram), Ensemble Methods (Bagging), t-SNE Visualization
Tools: Git, GitHub, LaTeX

ACHIEVEMENTS

Competitive Programming
Codeforces Expert (1602), CodeChef 5-Star (2022)

- Codeforces: Sabnaj_423 (Expert, Max 1602)
- CodeChef: Sabnaj42 (5-Star, Max 2022)

Programming Contest Rankings
National and Regional Programming Contests

- Intra KUET Programming Contest 2023: Ranked **2nd**
- National Girls' Programming Contest 2022: Ranked **8th**
- Ada Lovelace NGPC 2022: Ranked **10th**
- IUT IUPC 2024: Ranked **36th**
- ICPC Asia Dhaka Regional 2024: Ranked **61st**

LEADERSHIP & ACTIVITIES

Vice President **Feb 2024 – Feb 2025**
Special Group Interested in Programming Contest (SGIPC), KUET